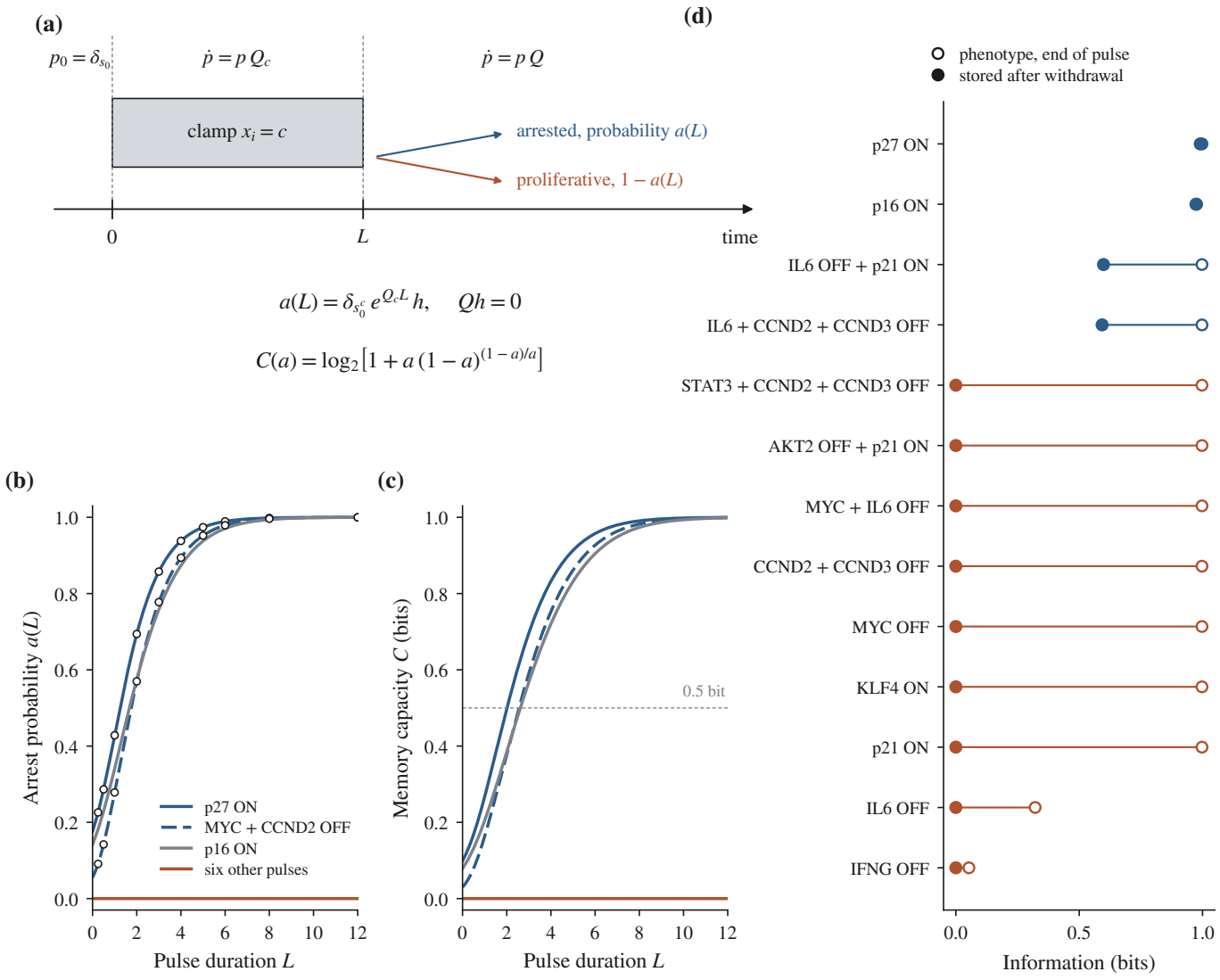
(a)
$p_0 = \delta_{s_0}$
$\dot{p} = p\,Q_c$
$\dot{p} = p\,Q$
clamp $x_i = c$
arrested, probability $a(L)$
proliferative, $1 - a(L)$
0
L
time
$a(L) = \delta_{s_0^c}\, e^{Q_c L}\, h, \quad Qh = 0$
$C(a) = \log_2\left[1 + a\,(1-a)^{(1-a)/a}\right]$
(b)
Arrest probability $a(L)$
Pulse duration L
p27 ON
MYC + CCND2 OFF
p16 ON
six other pulses
(c)
Memory capacity C (bits)
0.5 bit
Pulse duration L
(d)
phenotype, end of pulse
stored after withdrawal
p27 ON
p16 ON
IL6 OFF + p21 ON
IL6 + CCND2 + CCND3 OFF
STAT3 + CCND2 + CCND3 OFF
AKT2 OFF + p21 ON
MYC + IL6 OFF
CCND2 + CCND3 OFF
MYC OFF
KLF4 ON
p21 ON
IL6 OFF
IFNG OFF
Information (bits)